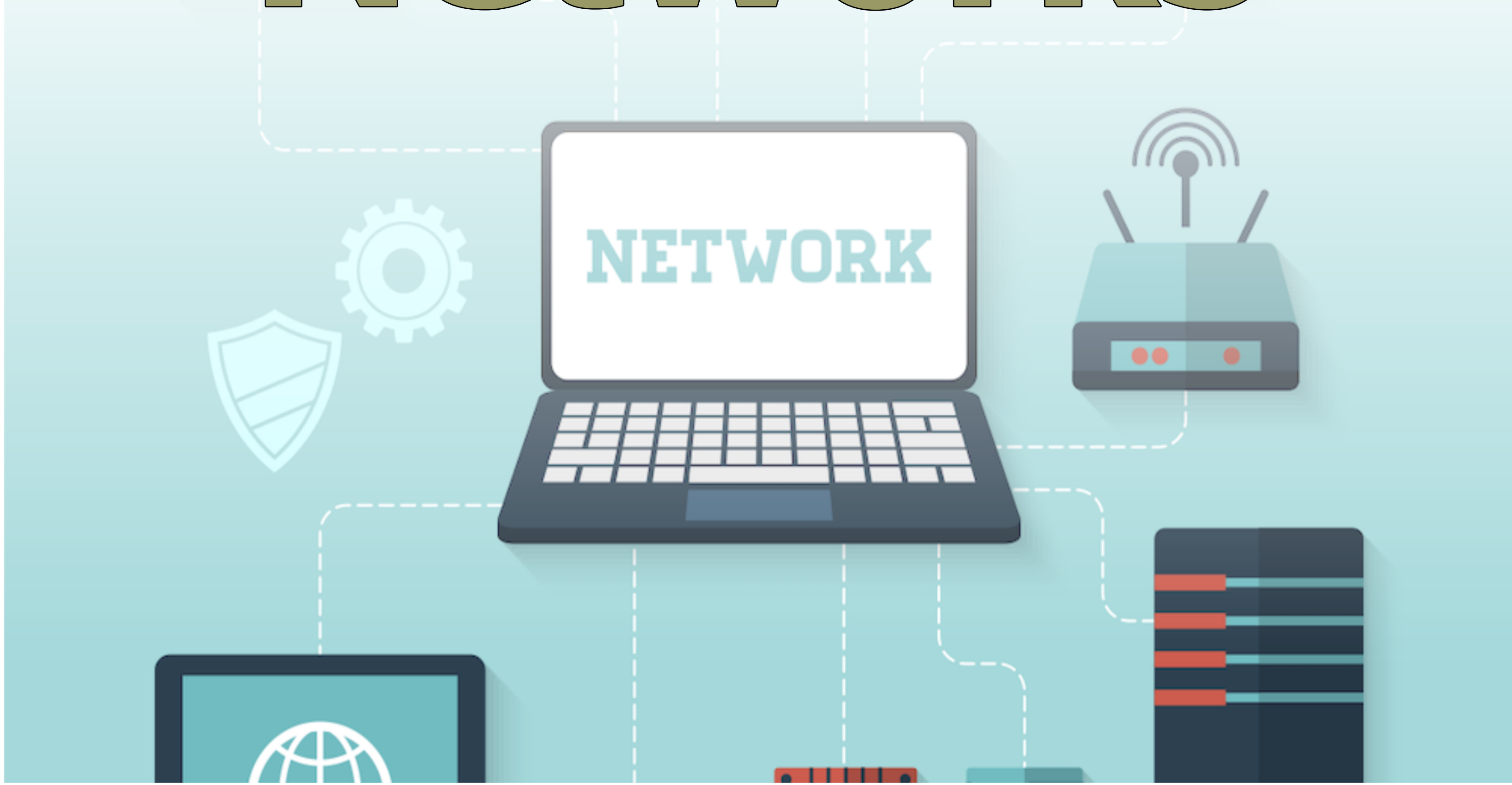
Computer Networks



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Mobile App: Physics Wallah (Available on Play Store)

Website: www.pw.live

Email: support@pw.live

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1 IP ADDRESSING, SUBNETTING & SUPERNETTING

1.1 IP Addressing

Class A : 0 → (1 - 126), No. of IP Addresses = 2^{31}
Class B : 10 → (128 - 191), No. of IP Addresses = 2^{30}
Class C : 110 → (192 - 223), No. of IP Addresses = 2^{29}
Class D : 1110 → (224 - 239), No. of IP Addresses = 2^{28}
Class E : 1111 → (240 - 255), No. of IP Addresses = 2^{28}

1.2 Default subnet Mask

Class A : 255.0.0.0
Class B : 255.255.0.0
Class C : 255.255.255.0

1.3 Private Addresses Range

10.0.0.0 to 10.255.255.255 → 1 class A Network.
172.16.0.0 to 172.31.255.255 → 16 class B Network.
192.168.0.0 to 192.168.255.255 → 256 class C Network.

Class	Number of Networks	Number of hosts per Network
Class A	$2^7 - 2 = 126$	$2^{24} - 2 = 1{,}67{,}77{,}214$ hosts
Class B	$2^{14} = 16{,}384$	$2^{16} - 2 = 65{,}534$ hosts
Class C	$2^{21} = 20{,}97{,}125$	$2^8 - 2 = 254$ hosts
Class D	No NID and HID, all 28 remaining bits are used to define multicast address	
Class E	No NID and HID, it is meant for research and future purpose	

Note:
The IP address 127.x.y.z is known as loop back address and it is used to check the connectivity.

1.4 Types of Communication

(i) Unicast communication (1 : 1)

(ii) Broadcast communication (1 : All)

(iii) Multicast Communication (1: Many)

1.5 Unicast Communication

1. Transmitting the data from one computer to another computer is called as unicast communication.
2. It is one to one transmission.
3. In Unicast communication both source and destination either present in the same network or in the different network.

1.6 Broadcast Communication

1.6.1 Limited Broadcasting

1. Transmitting data from one computer to all other computer in the same network is called as Limited Broadcasting.

2. Limited Broadcast Address = 255.255.255.255

3. Limited broadcast address can't be used as a source IP Address.

4. Limited broadcast Address will always be used as a Destination IP.

1.6.2 Direct Broadcasting

1. Transmitting data from one computer to all other computer in the different network is called as Limited Broadcasting.
2. Direct broadcast address can't be used as a source IP Address.
3. Direct broadcast Address will always be used as a Destination IP.

	NID	HID		
1.	–	0's	→	Network ID
2.	–	1's	→	Direct Broadcast Address (DBA)
3.	1's	1's	→	Limited Broadcast Address (LBA)
4.	0's	–	→	Host with in the Network
5.	1's	0's	→	Network Mask or Subnet Mask

1.7 Multicast communication

Transmitting a packet from one computer to many computers (0 or more) is called Multicast communication.

1.8 CIDR Rules

1. All the IP Address in the Block must be contiguous.
2. Block size must be a power of 2.
3. First IP address of the block must be divisible by size of the block.

1.9 Supernetting

The process of combining two or more network to get a single network is called as supernetting.

1.10 Advantage of Supernetting

1. Super netting Reduce Routing table entry.
2. Router will take less time for processing the packet.
3. It improve flexibility of IP Address Allotment i.e. If someone required 500 Address, then no need to purchase class B network we can combine two class C network.

1.11 Rules of Supernetting

1. Network ID must be contiguous.
2. Size of the Network must be same and number of Network must be a power of 2.
3. First Network ID must be divisible by size of the supernet.

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